

PROJECT REPORT ON

“ Z-BOT ”

Submitted by

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Under the Supervision of

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Submitted to

Tsinghua University for the award of

“Challenge Cup 2026”

Abstract

Scaffolding safety nets are widely used in high-rise construction to prevent falling debris, yet their deformable structure, debris accumulation, and concealed heat sources create significant inspection challenges. The Hong Kong scaffolding fire incident highlighted how combustible materials and hidden thermal hazards associated with scaffold systems can accelerate vertical fire spread. Current inspection methods rely on manual labor or aerial drones, which either expose workers to risk or lack stable close-contact sensing on compliant mesh surfaces.

This paper presents **Z-Bot**, a climbing robot specifically designed for deformable scaffolding net inspection. Unlike conventional wall-climbing robots that depend on suction or magnetic adhesion, Z-Bot employs a hook-based mechanical interlocking mechanism that directly engages mesh openings, enabling stable load transfer independent of surface continuity. The system integrates thermal sensing to detect concealed heat anomalies and a vision-based pipeline for debris identification.

Experimental evaluation includes climbing speed (cm/min), disturbance resistance under simulated wind conditions, and energy consumption over defined climbing distances. Results demonstrate stable locomotion and reliable hazard detection on compliant mesh structures, providing a safer and more proactive solution for construction site inspection.

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1 Introduction

In big cities today, we can see a lot of tall buildings being built. Because of that, construction sites are everywhere, and most of the time there is scaffolding around the outside of the building. Usually, workers install safety nets around the scaffolding. The main purpose is simple — to stop tools or materials from falling down and hurting people below. So these nets are actually very important.

But while learning more about this topic, I noticed that these nets can also cause problems. The net material is soft and flexible, not hard like a wall. When there is wind, it moves a lot. Since it stays outside for a long time, dust and trash can easily get stuck in it. After some time, the debris builds up. Some of this waste can burn easily, especially small plastic pieces or dry materials.

There have been some fire cases in Hong Kong where buildings under construction caught fire and the flames spread along the scaffolding area. From what I read, one reason could be the debris trapped in the safety nets. Also, during construction, workers use welding tools and temporary lighting. Electrical wires are also placed around the site. If something overheats or sparks behind the net, it may not be seen clearly from the ground. The net blocks part of the view.

Right now, inspection is mostly done manually. Workers need to climb up the scaffolding to check if everything is safe. This is obviously dangerous because they are working very high above the ground. Even though drones can help take videos, they cannot move the net or check behind it closely. So some hidden risks may still not be discovered.

There are wall-climbing robots that already exist, but most of them are made for solid surfaces like concrete or metal walls. They usually stick to the wall using suction or magnets. This works well on flat surfaces, but it doesn't really work on scaffolding nets because the net is soft and full of holes. The robot cannot attach properly.

Because of this, we came up with the idea of Z-Bot. Z-Bot is designed especially for climbing safety nets instead of solid walls. Instead of sticking to the surface, it uses small hooks to hold onto the holes in the mesh. This makes it more suitable for this kind of structure. We also added a thermal sensor to detect heat and a camera to observe debris. The idea is that it can help find possible fire hazards earlier.

In simple words, our goal is to make inspections safer. We want to reduce the need for workers to climb up the scaffolding just to check the nets. Z-Bot is designed to assist in this task and hopefully lower the risk of accidents on construction sit

2 Related Work

A. Wall-Climbing Robotic Systems

Wall-climbing robots are already used in many places. They help check bridges, building walls, and big tanks in factories. To climb, they need a way to stay attached to the surface.

Some robots use suction. This works well on smooth and flat walls. Some use magnets, but that only works if the wall is metal. Others use special sticky materials to grip the surface.

These methods are fine for solid walls. But they don't work well on soft or moving surfaces. If the surface has holes or keeps shaking, the robot cannot stay attached properly.

Scaffolding nets are very different from normal walls. They are soft. They have many holes. They move when there is wind. Because of that, most wall-climbing robots cannot work on them.

B. Moving on Mesh and Soft Surfaces

Climbing on a soft net is not easy. When weight is added, the net bends. This can make the robot unstable. If the wind blows, the net also moves. That makes it harder to control.

Most robots are made for hard surfaces. There is not much work on robots that climb mesh structures. In a mesh, there is no full surface. The robot can only hold onto small parts. This makes movement more difficult.

Also, at construction sites, dust and trash often get stuck in the net. This can block the holes or make the surface dirty. If a robot depends only on friction, it may slip.

So instead of using suction or sticky materials, Z-Bot uses small hooks. The hooks grab the holes in the net. It holds onto the structure itself. This makes it more stable, even if the net moves a little.

C. Inspection and Heat Detection

Construction sites can have fire risks. Wires can overheat. Welding can create sparks. If heat builds up somewhere, it should be found early.

Thermal sensors can detect high temperatures. Drones sometimes use thermal cameras to check buildings. They are useful, but they cannot get very close to the net. They also cannot look behind it easily. Wind can also affect them.

There are not many robots that can climb nets and check heat at the same time. Most climbing robots are still made for solid walls.

Z-Bot is designed for this kind of situation. It can climb the safety net using hooks. It also carries a thermal sensor and a camera. This way, it can check the net closely and look for possible fire risks.

That is the main idea behind it.

3 Requirements and Problem Definition

A. Problem Definition

Scaffolding nets on tall buildings are not easy for robots. They are soft and full of holes. They are not like walls. When something climbs on them, they bend. When there is wind, they move.

Because of this, it is hard for a robot to stay stable.

One problem is balance. The net does not stay still. If weight is added, it changes shape. Wind can make it shake more. So there is a higher chance of slipping.

Another problem is debris. Construction sites are usually dirty. Dust and small pieces of material get stuck in the net. Over time, more waste collects. Some of it can burn easily. It can also block the holes, which makes gripping harder.

There is also heat. Construction work uses wires, lights, and welding. These can create high temperatures. If something overheats behind the net, it may not be seen from the ground.

Right now, workers climb up to check the nets. This is risky because they are very high. Drones can take pictures, but they cannot touch the net or look closely behind it. Most climbing robots are made for hard walls, so they cannot work well on soft nets.

So the problem is clear. A robot is needed that can climb a soft net and also check for heat and debris.

B. Functional Requirements

- Z-Bot must be able to climb the net. It should grab the holes tightly and move up or sideways without falling.
- It should stay stable even if the net moves a little. It should not pull the net too much or damage it.
- It needs to detect heat. If there is a hot spot, it should notice it and send the information down.
- It also needs to see debris. A camera can check if too much waste is stuck in the net. The robot should save this information.
- It should not need someone to control every small movement. It should follow simple instructions and send data while working.

C. Other Requirements

- The robot must be safe. It should not fall easily. If one hook slips, the others should still hold.
- It should not use too much battery. It needs enough power to finish basic inspection.
- It must be lightweight. If it is too heavy, the net will stretch too much. It also needs to be small enough to fit between scaffold parts.
- Construction sites are dusty. The robot must still work in dust and changing light.

D. Testing

- Climbing speed can be checked by timing how long it takes to move up.
- Stability can be tested by shaking the net a little and seeing if it slips.
- Battery level can be checked before and after climbing.
- Heat detection can be tested by putting a heat source behind the net.
- Debris detection can be tested by placing waste in the net and seeing if the robot sees it.
- These tests show if the robot works or not.

E. Limits

- Not all nets are the same. Some are tighter. Some have bigger holes.
- Wind can make the net move more.
- The net cannot carry too much weight.

- The robot cannot carry heavy equipment.
- Lighting outside can change during the day.
- All of this affects how the robot is designed.

4 System Overview

4.1 Climbing “Pull-up” Mechanism”

4.2 Electrical Components

1. Raspberry pi 5
2. Arduino
3. Servo
4. Servo controller
5. Lipo Battery
6. Thermal Camera
7. Ordinary Camera

5 Mechanical Design

5.1 Limbs

The robot consists of 5 limbs in total: 3 front, and 2 back. Each joint is controlled by the same servo: HTD-45H Bus Serial Servo Motor 45kgcm Torque, which has a 240-degree range. All limbs are made from cuboid links, 3D printed using PLA, with the dimension 25×56×80mm.

5.1.1 Front Limb Type A

Its main purpose is for climbing. Located on the left and the right of the upper body
Hook–Servo1–Link1–Servo2–Servo3–Link2–Servo4–Body (see figure 1)

5.1.2 Front Limb Type B

Its main purpose is for stability. Located in front of the body.
Hook–Servo1–Link–Servo2–Body (see figure 2)



Figure 1. Front limb type A



Figure 2. Front limb type B

5.1.3 Back Limb

Its main purpose is for stability.

Hook–Servo1–Link–Servo2–Body (see figure 3)

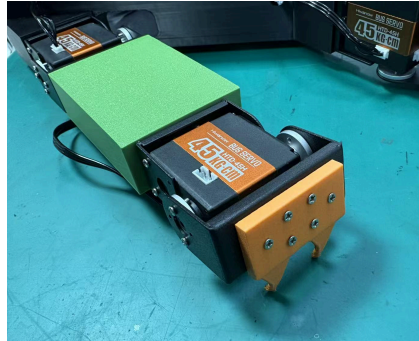


Figure 3. Back limb

5.2 Body

The body is flat and wide to keep the centre of mass near the net during climbing. All four “shoulders” point towards the front to help the limbs hook more reliably. (see figure 4)

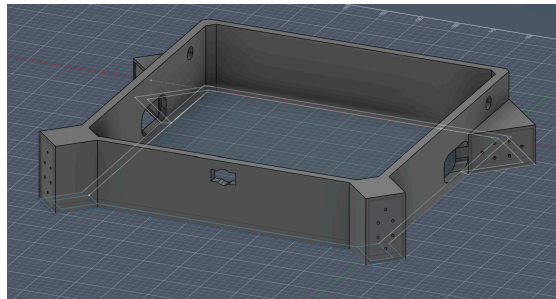


Figure 4. CAD design of the body

5.3 Hook

The hook is made using 3D printed PLA which is sufficiently strong for the robot. The hook is 3D printed so that the plane of the applied force is perpendicular to the grain(see figure 5).

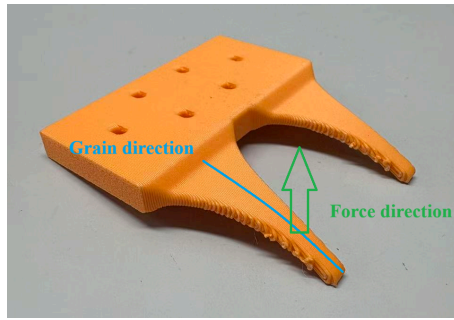


Figure 5. 3D printed hook

6 Modeling and Design Calculations

7 Electronics and Sensor Suite

8 Software Architecture and Algorithms

8.1 Control architecture

In control architecture, we mainly designed the Z-bot's movement mechanism based on the servo controllers. Servo controllers are practically and efficiently sufficient for net climbing as can be observed from the results in section 10. Servo controllers are adjusted and tuned through the **Bus Servo Controller**. 14 servo motors as explained in the mechanical design section are assigned their unique IDs and controlled in Bus Servo Controller. They are tuned based on the real physical tests for climbing up-down and left-right.

8.2 Locomotion control

The robot is adjusted to climb up-down and go left-right. The algorithm of all these processes are based on a sequential order of steps executed by servo controller.

8.2.1 Climb Up

8.2.1 Climb Down

8.3 Heat detection pipeline (behind net)

8.4 Debris detection pipeline

pretrained image segmentation algorithm + hsv color search on “background” segment

8.5 Large Language Model Vision

Input is fed into GLM 4.5 and turned to text via api

9 Prototype Iterations and Design Decisions

10 Experiments and Results

Types of climbing (speed/stability/energy) ! add simulations with Gemini Antigravity

Measure in cm/min

Shake net (wind) and see how easily they fall

Measure battery capacity after climbing n cm

11 Discussion

12 Conclusion and Future Work

13 References (nawat)